

Halfwave plate simulation

Divyesh Chauhan

What is a halfwave plate?

- ▶ It is an optical element that changes the relative phases difference between the x and y elements of the complex electric fields, effectively changing the one of the components' sign.
- ▶ Conventionally the lagging component is chosen to be given a the -ve sign.

Script 1: Energy density heat map I

- This python script creates an animation of the heatmap of the square of the real amplitude of the electric field.

```
\begin{lstlisting} def l_1 (z,v,lamd,t): v_x = v [0]* np.exp  
(1j2np.pi(ct-z)/lamd) v_y = v [1]* np.exp  
(1j2np.pi(ct-z)/lamd) return np.real (v_x)2 + np.real (v_y)2
```

```
def l_2 (z,v,lamd,phi_x,phi_y,t): v_x = v [0]* np.exp  
(1j2np.pi(ct-lamd*phi_x*z)/lamd) v_y = v [1]* np.exp  
(1j2np.pi(ct-lamd*phi_y*z)/lamd) return np.real (v_x)2 + np.real (v_y)2
```

```
def l_3 (z,v,d,lamd,phi_x,phi_y,t): v_x = v [0]* np.exp (   
1j2np.pi(ct-z+d)/lamd - phi_x*d ) v_y = v [1]* np.exp (   
1j2np.pi(ct-z+d)/lamd - phi_y*d ) return np.real (v_x)2 + np.real  
(v_y)2
```

Script 1: Energy density heat map II

```
def intensity (z,n_x,n_y,lamd,d,v,t): phi_x = 1j2np.pin_x/lamd  
phi_y = 1j2np.pin_y/lamd return (z<=0)I_1 (z,v,lamd,t) +  
np.logical_and (z>0,z<d)I_2 (z,v,lamd,phi_x,phi_y,t) + (z>=d)*  
I_3 (z,v,d,lamd,phi_x,phi_y,t) \end{lstlisting}
```